

Intermediate Code Generation

using Syntax-Directed Definitions (SDD)

and Syntax-Directed Translation (SDT)

Lecture Notes based on *Compilers: Principles, Techniques, and Tools* (the “Dragon Book”),
2nd Edition, Aho, Lam, Sethi, Ullman – Chapters 5 and 6

Suggested Lecture Plan (3–4 lectures)

- Lecture 1** Role of intermediate code; forms of IR: syntax trees, DAGs, three-address code, quadruples, triples, indirect triples.
- Lecture 2** *Syntax-directed definitions and translation, from scratch*: why syntax alone is not enough, attributes, semantic rules, synthesized and inherited attributes, dependency graphs, S- and L-attributed definitions; then SDTs: actions inside productions, postfix translation, marker nonterminals.
- Lecture 3** Translation of expressions and assignments, array references, type checking and conversion.
- Lecture 4** Boolean expressions and short-circuit code, flow-of-control statements, back-patching, switch statements and procedure calls; revision and exercises.

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1 Introduction: Why Intermediate Code?

In the *analysis-synthesis* model of a compiler, the **front end** analyses the source program and produces an **intermediate representation (IR)**; the **back end** synthesises the target program from that IR.

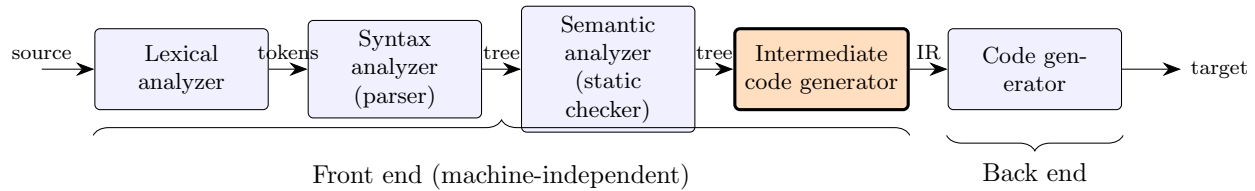


Figure 1: Position of the intermediate code generator in a compiler. The shaded (thick) box is the topic of these notes. In practice, static checking and intermediate code generation are often done together in one pass.

1.1 Advantages of using an intermediate representation

- **Retargeting.** A compiler for m languages and n machines needs only m front ends and n back ends (instead of $m \times n$ compilers) if they share one IR.
- **Machine-independent optimisation.** Optimisations (common sub-expression elimination, constant folding, loop-invariant code motion, ...) are applied once on the IR.
- **Simplicity.** Complex source constructs are lowered to a few simple operations, making code generation easier.

Static checking vs. intermediate code

Static checking includes type checking and syntactic checks that the grammar cannot enforce (e.g., **break** must be inside a loop). Dragon Book Chapter 6 treats type checking and intermediate-code generation together, because both are done by attaching attributes and actions to the grammar.

2 Variants of Syntax Trees and Intermediate Representations

2.1 Syntax trees (abstract syntax trees)

A **syntax tree** is a condensed form of the parse tree: operators and keywords appear as *interior nodes*, and operands as children. Chains of single productions disappear.

2.2 Directed acyclic graphs (DAGs) for expressions

A **DAG** identifies the *common sub-expressions*: a node is shared if the same sub-expression appears more than once. It shows the same information as a syntax tree but is more compact, and it directly suggests how to generate efficient code that computes a common sub-expression only once.

Example: tree vs. DAG for $a + a * (b - c) + (b - c) * d$

Figure 2 shows the syntax tree (left) and the DAG (right). In the DAG, the leaf a and the sub-expression $b - c$ are each shared by two parents.

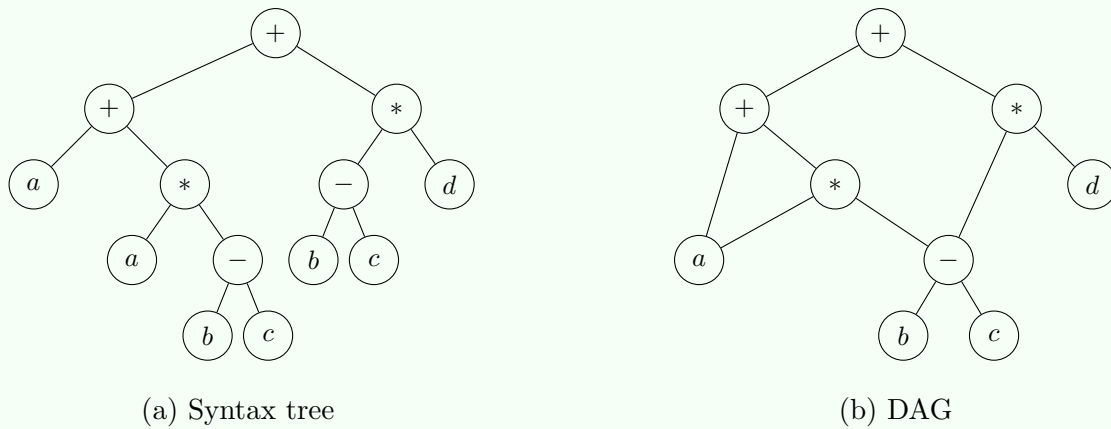


Figure 2: Syntax tree and DAG for $a + a * (b - c) + (b - c) * d$.

2.2.1 The value-number method for constructing DAGs

The problem. While building the tree for an expression, we want to *reuse* a node if the same sub-expression has already been built. To do this we need a fast way to answer: “Have I already created a node for $b - c$?” The **value-number method** gives a simple answer. Here we assume the *syntax tree* of the expression has already been built (Figure 2(a)). We convert it into a DAG by visiting the tree nodes in **post-order** (left subtree, right subtree, then the node itself), so that the children of a node are always handled before the node.

The idea. Keep all nodes in an **array of records**. Each record describes one node of the DAG, and the *position (index) of the record in the array is the node’s value number*. The value number is therefore just the node’s *name*: “node 4” means “the record at position 4”.

What the numbers mean

- **Value number** = the index of a record in the array = the identity of one DAG node. It is *not* the numeric value of the expression (the number 4 does not mean the expression evaluates to 4).
- A **leaf record** stores the kind of leaf and what it is: (**id**, pointer to symbol-table entry) for a variable, or (**num**, value) for a constant.
- An **interior record** has the form (*op*, *left*, *right*). Here *op* is the operator, and **left** and **right** are the **value numbers of the two child nodes**, i.e., they are *pointers* to other records (the operands).

Example. The record $(*, 1, 4)$ means: “a node for the operator $*$ whose left child is node 1 and whose right child is node 4”. Looking up the array, node 1 is the leaf a and node 4 is $b - c$; so this record stands for $a * (b - c)$.

Algorithm: finding or creating a node

To create a node with signature $\langle op, l, r \rangle$ (where l and r are value numbers of the children that were already created):

1. **Search** the array for a record with exactly the same op , $left = l$, and $right = r$.
2. If such a record **exists**, do not create anything; **return its value number**. (This is the common sub-expression being shared.)
3. Otherwise **append** a new record (op, l, r) at the end of the array and **return its index** as the new value number.

(For speed, a hash table keyed on (op, l, r) replaces the linear search in step 1.)

Why does comparing only (op, l, r) work? Two nodes represent the same expression exactly when they have the same operator and the same children. Children are the same exactly when they have the same value numbers (because every distinct sub-expression gets exactly one number). So, by building from the leaves upward, equal sub-expressions always get equal numbers.

Step-by-step example: $a + a * (b - c) + (b - c) * d$. We walk the syntax tree of Figure 2(a) in post-order: a , a , b , c , then $-$, $*$, $+$ of the left part, then b , c , $-$, d , $*$, and finally the root $+$. At every tree node we ask the array for the matching DAG node. Table 1 shows each request and what the algorithm does.

Step	Request	Found in array?	Action	Returns
1	leaf a	no	create record 1: (id, a)	1
2	leaf a (2nd time)	yes, record 1	nothing created	1
3	leaf b	no	create record 2: (id, b)	2
4	leaf c	no	create record 3: (id, c)	3
5	$\langle -, 2, 3 \rangle$ ($b - c$)	no	create record 4: $(-, 2, 3)$	4
6	$\langle *, 1, 4 \rangle$ ($a * (b - c)$)	no	create record 5: $(*, 1, 4)$	5
7	$\langle +, 1, 5 \rangle$ ($a + a * (b - c)$)	no	create record 6: $(+, 1, 5)$	6
8	leaf b (2nd time)	yes, record 2	nothing created	2
9	leaf c (2nd time)	yes, record 3	nothing created	3
10	$\langle -, 2, 3 \rangle$ ($b - c$ again)	yes, record 4	nothing created – <i>shared!</i>	4
11	leaf d	no	create record 7: (id, d)	7
12	$\langle *, 4, 7 \rangle$ ($(b - c) * d$)	no	create record 8: $(*, 4, 7)$	8
13	$\langle +, 6, 8 \rangle$ (whole expression)	no	create record 9: $(+, 6, 8)$	9

Table 1: Trace of the value-number method. Steps 2, 8, 9, 10 reuse existing nodes; step 10 is where the common sub-expression $b - c$ is detected.

The result has only 9 nodes (a plain syntax tree would have 13: it repeats a , b , c and the whole $b - c$ sub-tree). The final value number 9 identifies the root of the DAG.

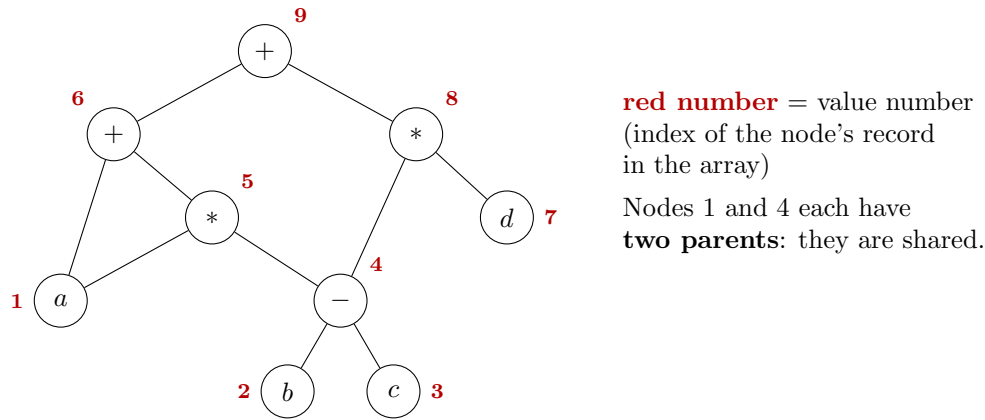


Figure 3: The DAG for $a + a * (b - c) + (b - c) * d$ with the value number of every node in red.

value no.	op	left	right
1	id <i>a</i>	–	–
2	id <i>b</i>	–	–
3	id <i>c</i>	–	–
4	–	2	3
5	*	1	4
6	+	1	5
7	id <i>d</i>	–	–
8	*	4	7
9	+	6	8

left = 1 points to record 1 (the leaf *a*)
right = 4 points to record 4 (the node *b - c*)
 So record 5 = (*, 1, 4) represents $a * (b - c)$.

Figure 4: The array of records after processing the example. Interior records contain *value numbers of their children*; here record 5 refers to records 1 and 4.

The whole procedure. Converting a syntax tree into a DAG is a single post-order traversal:

```

int buildDAG(TreeNode t) {           // returns a value number
    if (t is a leaf)
        return findOrCreateLeaf(t); // search (id/num, entry); create if absent
    l = buildDAG(t.left);             // children first (post-order)
    r = buildDAG(t.right);
    return findOrCreateNode(t.op, l, r); // search (op,l,r); create if absent
}
  
```

The “search” is only a *lookup of an identical record* in the array (or hash table); it is not a second traversal of the tree.

Looking ahead

Building the tree first and then converting it is easy to understand. Later, after we study syntax-directed definitions (Section 3.8), we will see that the same DAG can be built *directly while parsing*, without constructing the tree at all: the parser simply calls the find-or-create routine each time it recognises an operator.

2.3 Three-address code (TAC)

In **three-address code** every instruction has *at most one operator on the right side* and at most three addresses (two operands and one result). Compound expressions are broken into temporaries, so the order of evaluation is made explicit.

For $a + a * (b - c) + (b - c) * d$ the TAC (from the DAG, reusing $t_1 = b - c$) is:

```
t1 = b - c
t2 = a * t1
t3 = a + t2
t4 = t1 * d
t5 = t3 + t4
```

Without the DAG (from the tree) $b - c$ would be computed twice.

2.3.1 Generating three-address code from a DAG

Three-address code can be generated from the DAG alone; no SDD or SDT is needed.

Idea. Every *interior* node of the DAG becomes exactly *one* three-address instruction, with a fresh temporary holding its result. *Leaves* generate no instruction: they only supply a name (a variable such as a , or a constant). A shared node is still one node, so it produces only one instruction; this is how the DAG removes the common sub-expression.

Order. Process the records of the array in increasing order of value number. A node's children were always created before the node, so their value numbers are smaller, and their results are ready when the node is reached. The array order is therefore already a valid evaluation order.

```
for i = 1 to n:                                // records in the array, in index order
  if record i is a leaf:
    name[i] = the variable or constant it stands for
  else:
    // record i = (op, left, right)
    name[i] = newTemp()
    emit( name[i] = name[left] op name[right] )
```

Here $name[i]$ is the address that holds the value of node i . For an assignment $x = \langle \text{expression} \rangle$ we finish with one more instruction, $x = name[root]$.

Record	Content	$name[left]$	$name[right]$	$name[i]$	Instruction emitted
1	id a	–	–	a	(none, leaf)
2	id b	–	–	b	(none, leaf)
3	id c	–	–	c	(none, leaf)
4	$(-, 2, 3)$	b	c	t_1	$t_1 = b - c$
5	$(*, 1, 4)$	a	t_1	t_2	$t_2 = a * t_1$
6	$(+, 1, 5)$	a	t_2	t_3	$t_3 = a + t_2$
7	id d	–	–	d	(none, leaf)
8	$(*, 4, 7)$	t_1	d	t_4	$t_4 = t_1 * d$
9	$(+, 6, 8)$	t_3	t_4	t_5	$t_5 = t_3 + t_4$

Table 2: Generating three-address code from the array of Figure 4 (the DAG of $a + a * (b - c) + (b - c) * d$). Node 4 has two parents (nodes 5 and 8) but produces only one instruction; both parents use t_1 .

Three ways to obtain three-address code.

Method	How	Common sub-expressions
1. From a finished DAG	Scan the array in index order (above); no SDD/SDT needed.	Computed once (shared nodes).
2. During DAG construction	Whenever find-or-create <i>creates</i> a node, emit its instruction; when it finds an existing node, emit nothing. Needs SDT (Section 3.8).	Computed once.
3. SDD/SDT directly from the parse (Section 5.1)	Each reduction emits an instruction using <i>newTemp</i> and <i>gen</i> ; no DAG is built.	Computed again each time they occur (e.g., $b - c$ twice).

SDD/SDT is the *mechanism* for generating code in one pass while parsing; the DAG is what gives *sharing*. Methods 1 and 2 give the same code.

Caution: assignments inside the DAG

If the DAG covers several statements and a variable is *reassigned* in between (e.g., $a = a + 1$), a later use of a is a different value and must not share the old leaf or any node built from it. Compilers handle this by creating a new leaf (a new value number) after each assignment. For single expressions, as in these notes, the issue does not arise.

2.3.2 Addresses and instructions

An **address** can be: (i) a *name* (source variable; a pointer to its symbol-table entry), (ii) a *constant*, or (iii) a compiler-generated *temporary*.

Instruction form	Meaning
$x = y \text{ op } z$	binary assignment
$x = \text{op } y$	unary assignment (minus, not, conversions)
$x = y$	copy
goto L	unconditional jump
if x goto L / ifFalse x goto L	conditional jump on truth value
if x relop y goto L	conditional jump on comparison
param x, call p, n, y = call p, n, return y	procedure calls
$x = y[i], x[i] = y$	indexed copy (memory at address $y + i$)
$x = \&y, x = *y, *x = y$	address and pointer assignments
L:	label (a symbolic name for an instruction)

Table 3: Common three-address instruction forms.

2.3.3 Representations of three-address code

Take the source statement $a = b * -c + b * -c$.

(a) Three-address code

```

t1 = minus c
t2 = b * t1
t3 = minus c
t4 = b * t3
t5 = t2 + t4
a  = t5

```

(b) Quadruples – (op, arg1, arg2, result)

	op	arg ₁	arg ₂	result
0	minus	c		t1
1	*	b	t1	t2
2	minus	c		t3
3	*	b	t3	t4
4	+	t2	t4	t5
5	=	t5		a

(c) Triples – (op, arg1, arg2); a result is referred to by the *position* of its triple.

	op	arg ₁	arg ₂
0	minus	c	
1	*	b	(0)
2	minus	c	
3	*	b	(2)
4	+	(1)	(3)
5	=	a	(4)

(d) Indirect triples – a list of pointers to triples.

instruction	pointer
35	(0)
36	(1)
37	(2)
38	(3)
39	(4)
40	(5)

(triples as in (c))

	Advantage	Disadvantage
Quadruples	Results are named explicitly (temporaries); instructions can be moved freely during optimisation.	Needs extra space for temporaries and symbol-table entries.
Triples	No temporaries; more compact.	Moving a triple changes positions, so all references to it must be updated.
Indirect triples	Reordering only rearranges the pointer list, so triples keep their positions.	Extra level of indirection; an additional list is needed.

Table 4: Comparison of the three representations.

2.3.4 Static single-assignment (SSA) form – a brief note

In **SSA form** every variable is assigned exactly once; different assignments to the same source variable get different subscripts, and a ϕ -function merges values at control-flow joins:

p = a + b	p1 = a + b
q = p - c	q1 = p1 - c
p = q * d	p2 = q1 * d
p = e - p	p3 = e - p2
q = p + q	q2 = p3 + q1

```

if (flag) x = -1; else x = 1;  y = x * a;
==>  if (flag) x1 = -1; else x2 = 1;  x3 = phi(x1, x2);  y = x3 * a;

```

3 Syntax-Directed Definitions (SDD)

3.1 Why do we need them? Syntax is not enough

So far, parsing has answered only one question: “*Is this string a legal program, and what is its structure?*” A grammar and its parse tree describe **syntax**. They say nothing about **meaning** (*semantics*).

For example, the parse tree of $3 * 5 + 4$ shows that $3 * 5$ is grouped first, but nowhere does it contain the number 19. The parse tree of $a = b * c + d$ does not contain any machine or three-address instructions either. To translate a program we must *attach extra information to the nodes of the parse tree and compute it*. Typical questions are:

- What is the **value** of this expression? (calculators, constant folding)
- What is the **type** of this expression? (type checking)
- In which temporary or variable is its result **stored**? (code generation)
- Which **instructions** implement this statement? (code generation)
- To which **label** should control jump if this condition is true? (control flow)

A **syntax-directed definition** is the formal tool that answers such questions. The idea is simple:

**Grammar + extra information (attributes) on every grammar symbol
+ small rules (semantic rules) that say how to compute that information
from the production being used.**

“Syntax-directed” means that the computation is *driven by the syntactic structure* (the productions) of the input.

3.2 Attributes

An **attribute** is any value that we attach to a grammar symbol. It helps to picture the parse tree as a tree of *records*: every node has the grammar symbol as its name plus a few fields (the attributes). If X is a symbol and a is an attribute, we write $X.a$ for the value of a at a particular node labelled X . Attributes can be of any kind: numbers, strings, types, pointers to symbol-table entries, or even whole code sequences.

Attribute	Meaning	Typical symbols
$E.val$	numeric value of the expression	E, T, F
$E.type$	type of the expression (<code>int</code> , <code>float</code> , ...)	E
$id.lexeme$	the actual characters of the identifier (supplied by the lexer)	id
$digit.lexval$	numeric value of a digit token (supplied by the lexer)	digit , num
$E.addr$	name/temporary holding the result of E	E
$S.code$	three-address code for statement S	S, E, B
$B.true, B.false$	labels to jump to if B is true or false	B
$E.node$	pointer to the syntax-tree node for E	E, T

Different *occurrences* of the same nonterminal in a production are distinguished by subscripts, e.g., in $E \rightarrow E_1 + T$ the E on the left is the *head* and E_1 is the E in the body; both have a *val*.

3.3 Semantic rules: a first tiny example

A **semantic rule** is attached to a production and says how to compute one attribute from other attributes of the symbols in that same production. For the production $E \rightarrow E_1 + T$ the rule

$$E.val = E_1.val + T.val$$

says: “the value of the whole expression is the value of the left sub-expression plus the value of the term.” Consider the grammar $E \rightarrow E + T \mid T$, $T \rightarrow \mathbf{digit}$, together with the rules

Production	Semantic rule
$E \rightarrow E_1 + T$	$E.val = E_1.val + T.val$
$E \rightarrow T$	$E.val = T.val$
$T \rightarrow \mathbf{digit}$	$T.val = \mathbf{digit}.lexval$

Evaluate it on the input $2 + 3$. The parse tree is $E \Rightarrow E_1 + T$ with $E_1 \Rightarrow T \Rightarrow \mathbf{digit}(2)$ and $T \Rightarrow \mathbf{digit}(3)$. Working from the leaves upward:

1. The lexer supplies $\mathbf{digit}.lexval = 2$ and 3 .
2. $T \rightarrow \mathbf{digit}$ gives $T.val = 2$ (left) and $T.val = 3$ (right).
3. $E \rightarrow T$ gives $E_1.val = 2$.
4. $E \rightarrow E_1 + T$ gives $E.val = 2 + 3 = 5$.

That is all an SDD does: it decorates the parse tree with attribute values by repeatedly applying the rules.

3.4 Formal definition

Syntax-Directed Definition (SDD)

A **syntax-directed definition** is a context-free grammar together with **attributes** attached to grammar symbols and **semantic rules** attached to productions. For a production $A \rightarrow \alpha$ every rule has the form

$$b = f(c_1, c_2, \dots, c_k)$$

where f is a function and either

- (1) b is a **synthesized** attribute of A and $c_1, \dots, c_k$ are attributes of the symbols in α (the children) or of A itself; or
- (2) b is an **inherited** attribute of a symbol B in α and $c_1, \dots, c_k$ are attributes of A (the parent) or of symbols in α (the siblings), or of B itself.

3.5 Synthesized and inherited attributes

Analogy

Think of the parse tree as a company. A **synthesized** attribute is a *report sent up* to the manager: the manager’s value is built from the reports of the subordinates (e.g., the value of $2 + 3$ is built from the values of 2 and 3). An **inherited** attribute is an *instruction or information passed down or across*, from a parent or a left sibling (e.g., the declared type `float` is passed down to every identifier in `float a, b, c`).

- A **synthesized attribute** at a parse-tree node N is computed from attributes of N 's *children* (and N itself). Information flows *bottom-up*. Terminals have synthesized attributes only (supplied by the lexer, e.g., **digit.lexval**).
- An **inherited attribute** at N is computed from attributes of N 's *parent and siblings*. Information flows *top-down / sideways*. Useful when the structure of the parse tree does not match the abstract syntax, e.g., passing a type to a list of identifiers, or passing the label to jump to.

Example: synthesized attributes only (desk calculator)

Production	Semantic rules
$L \rightarrow E \mathbf{n}$	$L.val = E.val$
$E \rightarrow E_1 + T$	$E.val = E_1.val + T.val$
$E \rightarrow T$	$E.val = T.val$
$T \rightarrow T_1 * F$	$T.val = T_1.val \times F.val$
$T \rightarrow F$	$T.val = F.val$
$F \rightarrow (E)$	$F.val = E.val$
$F \rightarrow \mathbf{digit}$	$F.val = \mathbf{digit.lexval}$

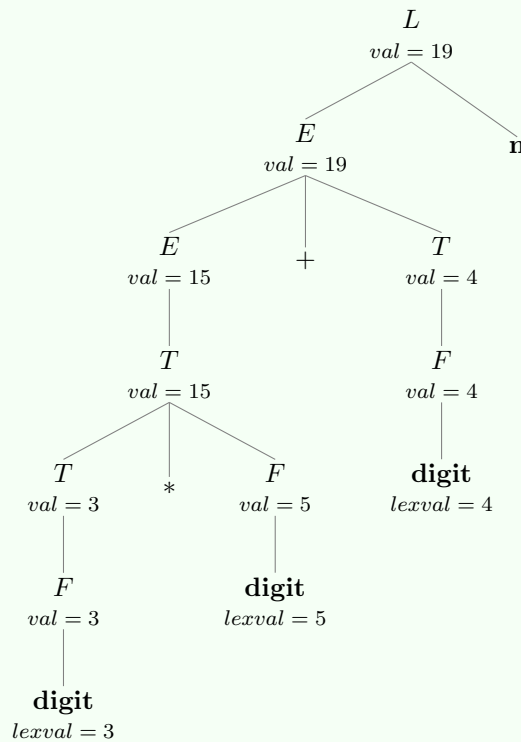


Figure 5: Annotated parse tree for $3*5+4 \mathbf{n}$. Each node shows the value of its synthesized attribute val .

How to read the figure (evaluation recipe).

1. The lexer supplies the $lexval$ of every **digit** leaf (3, 5, 4).
2. Each $F \rightarrow \mathbf{digit}$ copies it into $F.val$; each $T \rightarrow F$ copies it up again.
3. $T \rightarrow T_1 * F$ is applied once both its children are known: $3 \times 5 = 15$.
4. $E \rightarrow E_1 + T$ then adds $15 + 4 = 19$, and $L \rightarrow E \mathbf{n}$ passes 19 to the root.

Every node is computed *after* its children, i.e., in **post-order**. This is natural for synthesized attributes, and it is exactly the order in which an LR parser performs reductions.

3.6 Annotated parse trees and dependency graphs

An **annotated parse tree** shows the attribute values at every node. To evaluate attributes we build a **dependency graph**: one node per attribute instance, and an edge $x \rightarrow y$ whenever y is computed from x . A valid evaluation order is any *topological order* of this graph; if the graph has a cycle, no order exists.

A circular definition

For $A \rightarrow B$ with the rules $A.s = B.i + 1$ and $B.i = A.s \times 2$, the value of $A.s$ needs $B.i$, which needs $A.s$. The dependency graph has a *cycle*, so no evaluation order exists. A good SDD must be *non-circular*; the classes S-attributed and L-attributed below guarantee this automatically.

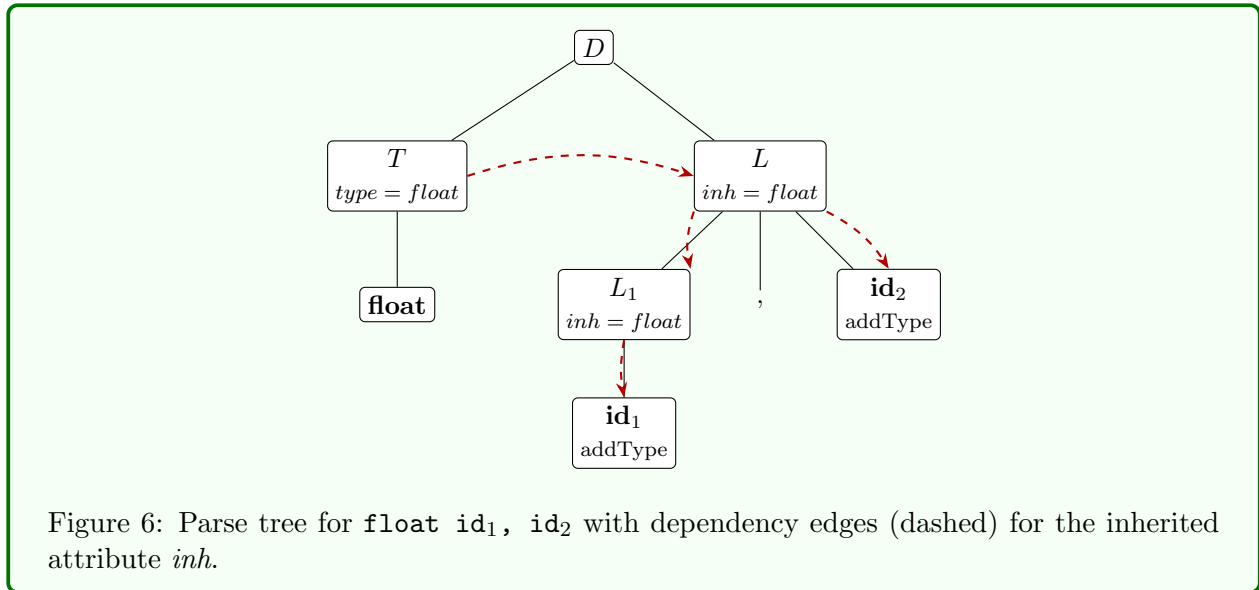
Side effects

A semantic rule may also do something besides computing a value, e.g., `addType(entry, t)` writes into the symbol table, `print` writes to the output, and `gen` emits an instruction. Such rules are called **side effects**; they are fine as long as the order in which they run does not change the result (or the order is controlled, which is the job of an SDT).

Example: inherited attributes (declaration list)

Production	Semantic rules
$D \rightarrow T L$	$L.inh = T.type$
$T \rightarrow \text{int}$	$T.type = integer$
$T \rightarrow \text{float}$	$T.type = float$
$L \rightarrow L_1, \text{id}$	$L_1.inh = L.inh; \text{addType}(\text{id.entry}, L.inh)$
$L \rightarrow \text{id}$	$\text{addType}(\text{id.entry}, L.inh)$

For the input `float id1, id2` the dashed arrows in Fig. 6 are the dependency-graph edges. The type is passed *down and across* from T to L , then down through the list to each identifier.



3.7 Classes of SDDs that are easy to implement

S-attributed and L-attributed definitions

- An SDD is **S-attributed** if *every attribute is synthesized*. It can be evaluated in any bottom-up order, e.g., during LR parsing, at the time of reduction.
- An SDD is **L-attributed** if each attribute is either synthesized, or inherited such that, for a production $A \rightarrow X_1X_2 \cdots X_n$, the inherited attribute of X_i depends only on:
 - (a) inherited attributes of A , and
 - (b) attributes (inherited or synthesized) of $X_1, \dots, X_{i-1}$ (the symbols to its *left*), and
 - (c) attributes of X_i itself, provided there is no cycle.

Evaluation is possible in a single left-to-right depth-first traversal (e.g., during top-down/LL parsing).

Every S-attributed SDD is also L-attributed.

Not L-attributed

For $A \rightarrow BC$, the rule $C.i = B.s$ is fine (the inherited attribute of C uses its *left* sibling B). But the rule $B.i = f(C.x)$ makes the inherited attribute of B depend on its *right* sibling C , so the SDD is *not* L-attributed.

3.8 SDD for constructing syntax trees

Syntax trees can be built using synthesized attribute `node`, which points to the root of a (sub)tree, and constructor functions $Node(op, c_1, \dots)$ and $Leaf(id, entry) / Leaf(num, val)$.

Production	Semantic rules
$E \rightarrow E_1 + T$	$E.node = \mathbf{new} \text{ Node}('+', E_1.node, T.node)$
$E \rightarrow E_1 - T$	$E.node = \mathbf{new} \text{ Node}('-', E_1.node, T.node)$
$E \rightarrow T$	$E.node = T.node$
$T \rightarrow (E)$	$T.node = E.node$
$T \rightarrow \mathbf{id}$	$T.node = \mathbf{new} \text{ Leaf}(\mathbf{id}, \mathbf{id.entry})$
$T \rightarrow \mathbf{num}$	$T.node = \mathbf{new} \text{ Leaf}(\mathbf{num}, \mathbf{num.val})$

Building a DAG directly. If each $\mathbf{new} \text{ Node}(\dots)$ / $\mathbf{new} \text{ Leaf}(\dots)$ is replaced by the find-or-create routine of Section 2.2.1 (which returns an existing node when an identical one exists), the same SDD builds a DAG during parsing, and $E.node$ holds a value number.

This is S-attributed. If the grammar is made LL by eliminating left recursion, the same tree is built with an inherited attribute (see the L-attributed SDT in Section 4.6).

4 Syntax-Directed Translation Schemes (SDT)

4.1 From specification to implementation

An SDD tells us *what* every attribute must equal, but it does not say *when* each rule should run. A real compiler, however, runs the rules *while it is parsing*, in one definite order, and often without ever building the whole parse tree. A **syntax-directed translation scheme (SDT)** solves this: it takes the program fragments of the rules and writes them *inside the production body, at the exact position where they should run*.

The same rule written both ways:

SDD (specification)	SDT (implementation)
$E \rightarrow E_1 + T$ rule: $E.val = E_1.val + T.val$	$E \rightarrow E_1 + T \{ E.val = E_1.val + T.val \}$

	SDD	SDT
Nature	High-level specification: rules attached to productions	Concrete implementation: actions embedded in productions
Order of evaluation	Not specified (any order consistent with dependencies)	Explicit: actions run in the left-to-right order of the production body
Rules/actions	Written next to the production	Written in braces $\{\dots\}$ inside the body
Used for	Designing and reasoning about the translation	Building the actual translator

4.2 Definition and the meaning of an action's position

Syntax-Directed Translation Scheme

A **syntax-directed translation scheme (SDT)** is a context-free grammar with **program fragments** (*semantic actions*) embedded within the production bodies, written in braces.

The position of an action defines *when* it is executed during parsing.

$$B \rightarrow X \{a\} Y$$

Here action a is executed after X has been recognised (and before Y).

How an SDT is executed. Imagine the parse tree in which every action appears as an *extra child*, exactly at the position where it is written in the production. Perform a **depth-first traversal from left to right**; whenever the traversal reaches an action, it is executed. Figure 7 shows this for a scheme that translates infix to postfix: $\text{expr} \rightarrow \text{expr} + \text{term} \{\text{print('+')}\} \mid \text{expr} - \text{term} \{\text{print('-')}\} \mid \text{term}$ and $\text{term} \rightarrow \text{digit} \{\text{print(digit)}\}$.

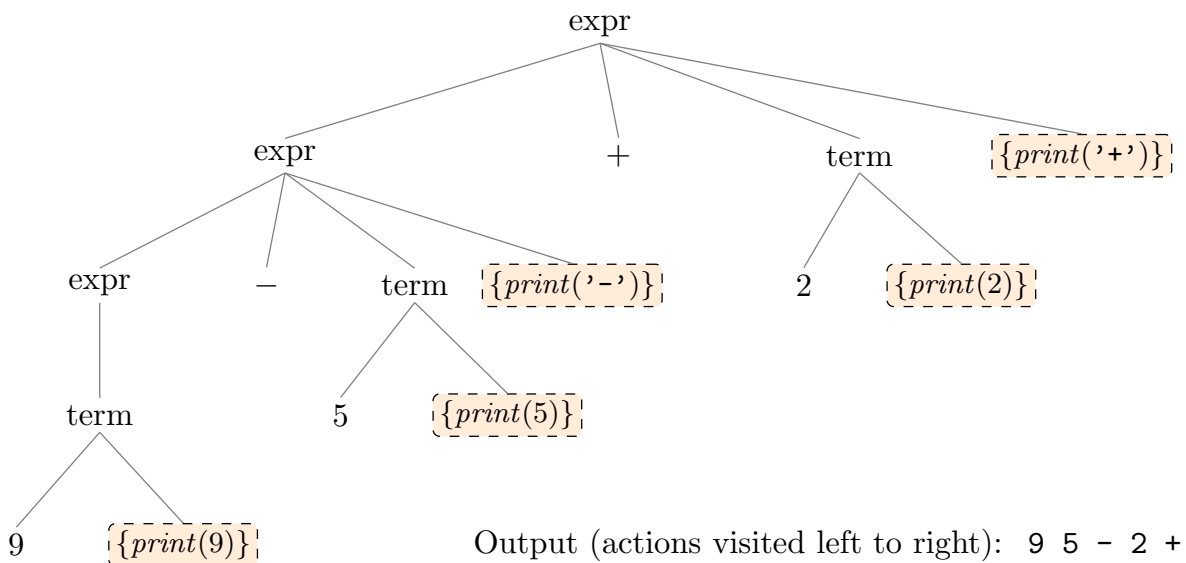


Figure 7: Parse tree of $9-5+2$ with actions (dashed boxes) as extra children. Visiting the tree depth-first, left to right, executes the actions in the order 9, 5, -, 2, +, producing the postfix form.

In practice the parser does not need to build this tree: a bottom-up parser executes an action when the symbols to its left have been reduced, and a top-down parser executes it at the corresponding point of the recursive-descent procedure.

4.3 Postfix translation schemes

If the grammar is LR-parsable and the SDD is **S-attributed**, we can build an SDT with every action at the *right end* of its production (a **postfix SDT**). The actions then execute at *reduction time*, so the attribute of each grammar symbol can be stored beside it on the parser stack.

Postfix SDT for the desk calculator

$L \rightarrow E \mathbf{n}$	$\{ \text{print}(E.val) \}$
$E \rightarrow E_1 + T$	$\{ E.val = E_1.val + T.val \}$
$E \rightarrow T$	$\{ E.val = T.val \}$
$T \rightarrow T_1 * F$	$\{ T.val = T_1.val \times F.val \}$
$T \rightarrow F$	$\{ T.val = F.val \}$
$F \rightarrow (E)$	$\{ F.val = E.val \}$
$F \rightarrow \mathbf{digit}$	$\{ F.val = \mathbf{digit.lexval} \}$

Implementation on the LR parse stack. Each stack entry holds a grammar symbol and its attribute value. When $A \rightarrow XYZ$ is reduced, the actions read the values in the top three entries ($\text{stack}[\text{top}-2].val$, $\text{stack}[\text{top}-1].val$, $\text{stack}[\text{top}].val$), compute $A.val$, pop three entries, and push A with the new value.

Stack (symbol:val)	Remaining input	Action
\$	3 * 5 + 4 n	shift
\$ digit :3	* 5 + 4 n	reduce $F \rightarrow \mathbf{digit}$
\$ F :3	* 5 + 4 n	reduce $T \rightarrow F$
\$ T :3	* 5 + 4 n	shift *, shift digit
\$ T :3 * digit :5	+ 4 n	reduce $F \rightarrow \mathbf{digit}$
\$ T :3 * F :5	+ 4 n	reduce $T \rightarrow T * F$ ($3 \times 5 = 15$)
\$ T :15	+ 4 n	reduce $E \rightarrow T$
\$ E :15	+ 4 n	shift +, shift digit , reduce to T :4
\$ E :15 + T :4	n	reduce $E \rightarrow E + T$ ($15 + 4 = 19$)
\$ E :19	n	shift n , reduce $L \rightarrow E \mathbf{n}$, print 19

4.4 SDTs with actions inside productions

When an action is in the middle of a production, it is performed *immediately after all grammar symbols to its left have been matched*. In a top-down parser the action is executed at that point of the recursive-descent procedure; in a bottom-up parser it is performed right after the symbols to its left are on the stack.

Example: infix to postfix (action placed after the operand)

$E \rightarrow T R$
$R \rightarrow + T \{ \text{print}(' + ') \} R_1 \mid - T \{ \text{print}(' - ') \} R_1 \mid \varepsilon$
$T \rightarrow \mathbf{num} \{ \text{print}(\mathbf{num.val}) \}$

For 9-5+2 the output is 95-2+.

4.5 Eliminating left recursion from SDTs and marker nonterminals

Marker nonterminals (for bottom-up parsing). Suppose an action must appear in the middle of a production, e.g., $B \rightarrow X \{a\} Y$. For an LR parser we replace the action by a **marker nonterminal** M with an empty production:

$$B \rightarrow X M Y, \quad M \rightarrow \varepsilon \{a\}$$

The action a is then executed upon reducing $M \rightarrow \varepsilon$. (This can introduce conflicts if the grammar is not careful; the marker must not cause a parsing-action conflict.) We will use markers M and N extensively in backpatching (Section 8).

4.6 Converting an L-attributed SDD into an SDT

Rules for converting an L-attributed SDD to an SDT

1. **Inherited attribute** of a nonterminal X in the body: embed the action that computes it *immediately before* X in the body. If several inherited attributes of X depend on one another, order the actions appropriately.
2. **Synthesized attribute** of the head A : place the action that computes it at the *end* of the production body.

Example: syntax tree for a left-recursion-free grammar

After removing left recursion from $E \rightarrow E_1 + T \mid E_1 - T \mid T$:

$$\begin{aligned} E &\rightarrow T \{ R.i = T.node \} R \{ E.node = R.s \} \\ R &\rightarrow + T \{ R_1.i = \text{new Node}(' + ', R.i, T.node) \} R_1 \{ R.s = R_1.s \} \\ R &\rightarrow - T \{ R_1.i = \text{new Node}(' - ', R.i, T.node) \} R_1 \{ R.s = R_1.s \} \\ R &\rightarrow \varepsilon \{ R.s = R.i \} \\ T &\rightarrow \text{id} \{ T.node = \text{new Leaf}(\text{id}, \text{id.entry}) \} \end{aligned}$$

Here $R.i$ (inherited) carries the tree built *so far*; $R.s$ (synthesized) returns the final tree. For $a - 4 + c$ the tree $((a - 4) + c)$ is produced with the correct left-associativity.

5 Translation of Expressions and Assignments

5.1 SDD for three-address code

We use the following:

- $E.addr$: the address (name, constant, or temporary) that holds the value of E .
- $E.code$ (or $S.code$): the three-address instructions that evaluate E (or execute S).
- $newTemp()$ (written **new** $Temp$): returns a fresh temporary $t_1, t_2, \dots$ in successive calls.
- **gen**($\dots$): builds a three-address instruction. $\parallel$ denotes concatenation of instruction sequences.

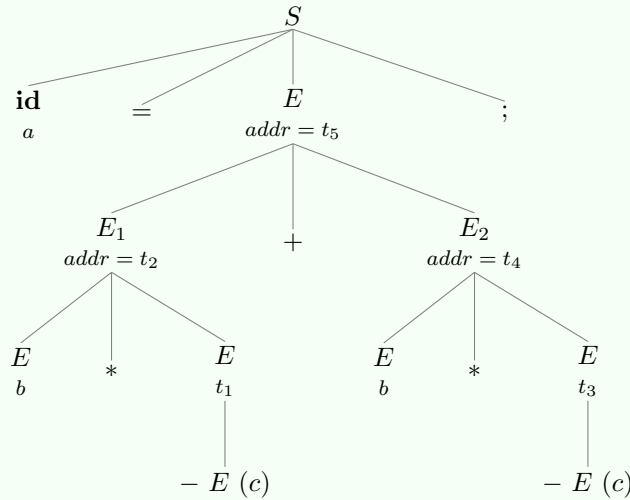
Production	Semantic rules
$S \rightarrow \text{id} = E ;$	$S.code = E.code \parallel \text{gen}(top.get(\text{id.lexeme}) = E.addr)$
$E \rightarrow E_1 + E_2$	$E.addr = \text{new Temp}()$ $E.code = E_1.code \parallel E_2.code \parallel \text{gen}(E.addr = E_1.addr + E_2.addr)$
$E \rightarrow - E_1$	$E.addr = \text{new Temp}()$ $E.code = E_1.code \parallel \text{gen}(E.addr = \text{minus } E_1.addr)$
$E \rightarrow (E_1)$	$E.addr = E_1.addr; \quad E.code = E_1.code$
$E \rightarrow \text{id}$	$E.addr = top.get(\text{id.lexeme}); \quad E.code = ''$

Table 5: SDD to produce three-address code for expressions and assignments (Dragon Book Fig. 6.19).

Remark

Here *top* is the current symbol table; *top.get(name)* returns the symbol-table entry. The code of a node is the *concatenation* of its children's code followed by one new instruction, so the instructions are produced in postfix (evaluation) order. The *.code* attribute is a long string, so this SDD is inefficient in practice, which motivates the incremental approach below.

Worked example: $a = b * -c + b * -c ;$



Concatenating the code in the order of evaluation gives:

```

t1 = minus c
t2 = b * t1
t3 = minus c
t4 = b * t3
t5 = t2 + t4
a = t5

```

5.2 Incremental translation (SDT with gen emitting directly)

Building *E.code* as a string copies code repeatedly. In an *incremental* scheme, *gen()* **appends** the instruction to an output stream/array as soon as the action is executed. The code attribute is not stored. Since actions run in the correct order (postfix), the instructions come out in the right order.

Production	Translation scheme (actions in braces)
$S \rightarrow \mathbf{id} = E ;$	$\{ \text{gen}(\text{top.get}(\mathbf{id.lexeme}) = E.addr); \}$
$E \rightarrow E_1 + E_2$	$\{ E.addr = \mathbf{new Temp}(); \text{gen}(E.addr = E_1.addr + E_2.addr); \}$
$ -E_1$	$\{ E.addr = \mathbf{new Temp}(); \text{gen}(E.addr = \mathbf{minus } E_1.addr); \}$
$ (E_1)$	$\{ E.addr = E_1.addr; \}$
$ \mathbf{id}$	$\{ E.addr = \text{top.get}(\mathbf{id.lexeme}); \}$

Table 6: Incremental SDT for expressions (all actions are at the right end, so it is a postfix SDT, suitable for LR parsing).

5.3 Addressing array elements

Let an array be stored in a contiguous block of memory with base address *base* and element width *w*.

One-dimensional array $A[\text{low} \dots \text{high}]$:

$$\text{address of } A[i] = \text{base} + (i - \text{low}) \times w$$

If *low* = 0 this is simply $\text{base} + i \times w$.

Two-dimensional array (row-major) $A[n_1][n_2]$ with element width *w*:

$$\text{address of } A[i_1][i_2] = \text{base} + (i_1 \times n_2 + i_2) \times w$$

***k* dimensions** $A[n_1] \dots [n_k]$ (row-major, 0-based):

$$\text{base} + ((\dots ((i_1 n_2 + i_2) n_3 + i_3) \dots) n_k + i_k) \times w$$

Equivalently, using the *widths of sub-arrays* $w_1 = n_2 \dots n_k \cdot w$, $w_2 = n_3 \dots n_k \cdot w, \dots, w_k = w$:

$$\text{base} + i_1 \times w_1 + i_2 \times w_2 + \dots + i_k \times w_k.$$

This form is the one used in the translation scheme.

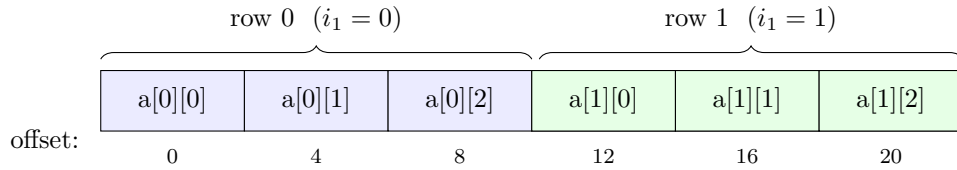


Figure 8: Row-major layout of `int a[2][3]` ($w = 4$). Offset of $a[i][j]$ is $i \times 12 + j \times 4$.

5.4 Translation of array references

Grammar (with nonterminal *L* for an array location):

$$\begin{aligned}
 S &\rightarrow \mathbf{id} = E ; \mid L = E ; \\
 E &\rightarrow E_1 + E_2 \mid \mathbf{id} \mid L \\
 L &\rightarrow \mathbf{id} [E] \mid L_1 [E]
 \end{aligned}$$

Attributes of *L*: *L.array* (symbol-table entry of the array), *L.type* (type of the *sub-array* generated by *L*), *L.addr* (a temporary accumulating the *offset*).

Production	Semantic action
$S \rightarrow \text{id} = E ;$	$\{ \text{gen}(\text{top.get}(\text{id.lexeme}) = E.\text{addr}); \}$
$S \rightarrow L = E ;$	$\{ \text{gen}(L.\text{array.base} [L.\text{addr}] = E.\text{addr}); \}$
$E \rightarrow E_1 + E_2$	$\{ E.\text{addr} = \text{new Temp}(); \text{gen}(E.\text{addr} = E_1.\text{addr} + E_2.\text{addr}); \}$
$E \rightarrow \text{id}$	$\{ E.\text{addr} = \text{top.get}(\text{id.lexeme}); \}$
$E \rightarrow L$	$\{ E.\text{addr} = \text{new Temp}(); \text{gen}(E.\text{addr} = L.\text{array.base} [L.\text{addr}]); \}$
$L \rightarrow \text{id} [E]$	$\{ L.\text{array} = \text{top.get}(\text{id.lexeme}); L.\text{type} = L.\text{array.type.elem};$ $L.\text{addr} = \text{new Temp}(); \text{gen}(L.\text{addr} = E.\text{addr} * L.\text{type.width}); \}$
$L \rightarrow L_1 [E]$	$\{ L.\text{array} = L_1.\text{array}; L.\text{type} = L_1.\text{type.elem};$ $t = \text{new Temp}(); L.\text{addr} = \text{new Temp}();$ $\text{gen}(t = E.\text{addr} * L.\text{type.width}); \text{gen}(L.\text{addr} = L_1.\text{addr} + t); \}$

Table 7: Semantic actions for array references (Dragon Book Fig. 6.22).

Worked example: $c + a[i][j]$ where a is `int a[2][3]`

Types: $a.\text{type} = \text{array}(2, \text{array}(3, \text{integer}))$, so the width of one row is $3 \times 4 = 12$ and of an element is 4.

1. $L \rightarrow \text{id}[E]$ with $E = i$: $L.\text{type} = \text{array}(3, \text{integer})$, width 12 $\Rightarrow t1 = i * 12$.
2. $L \rightarrow L_1[E]$ with $E = j$: $L.\text{type} = \text{integer}$, width 4 $\Rightarrow t2 = j * 4, t3 = t1 + t2$.
3. $E \rightarrow L$: $t4 = a[t3]$.
4. $E \rightarrow E_1 + E_2$: $t5 = c + t4$.

```
t1 = i * 12
t2 = j * 4
t3 = t1 + t2
t4 = a[t3]
t5 = c + t4
```

And for the assignment $x = a[i][j]$ the code is the first four instructions followed by $x = t4$.

6 Type Checking and Type Conversion

Type expressions represent the type of a language construct: basic types (`int`, `float`, `char`, `boolean`, `void`), $\text{array}(n, T)$, *record*, pointer, and function types $s \rightarrow t$.

Type checking comes in two forms: **synthesis** (the type of an expression is built from the types of its sub-expressions, e.g., for $E_1 + E_2$) and **inference** (the type of a construct is determined from the way it is used).

6.1 Type conversion (coercion)

For $E \rightarrow E_1 + E_2$ where operand types may differ (e.g., `int` and `float`), the compiler inserts conversions. Define the widening hierarchy $\text{int} < \text{float}$. Then

$$E.\text{type} = \max(E_1.\text{type}, E_2.\text{type}),$$

and the helper $\text{widen}(a, t, w)$ converts address a of type t to type w :

```
Addr widen(Addr a, Type t, Type w) {
```

```

    if (t == w) return a;
    else if (t == integer && w == float) {
        temp = new Temp();
        gen(temp '=' '(float)' a);
        return temp;
    }
    else error;
}

```

Production	Semantic action
$E \rightarrow E_1 + E_2$	$\{ E.type = \max(E_1.type, E_2.type);$ $a_1 = \text{widen}(E_1.addr, E_1.type, E.type);$ $a_2 = \text{widen}(E_2.addr, E_2.type, E.type);$ $E.addr = \text{new Temp}(); \text{ gen}(E.addr = a_1 + a_2); \}$

Example

Let x be float, and i, j be int. For $x = i * j + x$ we get:

```

t1 = i * j      // int * int -> int
t2 = (float) t1 // widen to float
t3 = t2 + x     // float + float
x  = t3

```

7 Control Flow: Boolean Expressions and Statements

In programming languages, boolean expressions are used for two purposes:

1. to **compute logical values** (e.g., `flag = a < b && c;`), and
2. as **conditional expressions** in `if`, `while`, and `for` (flow-of-control).

We use the grammar

$$B \rightarrow B \parallel B \mid B \&\& B \mid !B \mid (B) \mid E \text{ rel } E \mid \text{true} \mid \text{false}$$

with $\parallel$ standing for the C operator `||`; `||` and `&&` are left-associative, `||` has lower precedence than `&&`, which is lower than `!`.

7.1 Short-circuit (jumping) code

In **short-circuit code**, the operators `&&`, `||` and `!` do not appear in the generated code. Instead a boolean expression is translated into jumps to labels: control *reaches* one position if the expression is true and another if it is false. In

```
if (x < 100 || x > 200 && x != y) x = 0;
```

the first test `x<100` being true makes the other tests unnecessary. The generated code is:

```

    if x < 100 goto L2
    goto L3
L3:    if x > 200 goto L4
    goto L1
L4:    if x != y goto L2

```

```

        goto L1
L2:    x = 0
L1:    ...

```

(Compare with *value* code, which would compute 0/1 into temporaries and use `&&` and `||` at run time.)

7.2 Flow-of-control statements: SDD with inherited labels

Grammar for statements:

$$P \rightarrow S, \quad S \rightarrow \text{assign} \mid \text{if}(B) S_1 \mid \text{if}(B) S_1 \text{ else } S_2 \mid \text{while}(B) S_1 \mid S_1 S_2$$

Inherited attributes (labels):

- $B.true$: the label to jump to if B is true; $B.false$: the label if B is false.
- $S.next$: the label of the instruction that follows the code for S .

Synthesized attribute: $S.code$, $B.code$. The function $newlabel()$ creates a fresh label $L_1, L_2, \dots$, and $label(L)$ attaches label L to the next instruction.

Production	Semantic rules
$P \rightarrow S$	$S.next = newlabel(); \quad P.code = S.code \parallel label(S.next)$
$S \rightarrow \text{assign}$	$S.code = \text{assign}.code$
$S \rightarrow \text{if}(B) S_1$	$B.true = newlabel(); \quad B.false = S_1.next = S.next$ $S.code = B.code \parallel label(B.true) \parallel S_1.code$
$S \rightarrow \text{if}(B) S_1 \text{ else } S_2$	$B.true = newlabel(); \quad B.false = newlabel(); \quad S_1.next = S_2.next = S.next$ $S.code = B.code \parallel label(B.true) \parallel S_1.code \parallel \text{gen}(\text{goto } S.next) \parallel label(B.false) \parallel S_2.code$
$S \rightarrow \text{while}(B) S_1$	$begin = newlabel(); \quad B.true = newlabel(); \quad B.false = S.next;$ $S_1.next = begin$ $S.code = label(begin) \parallel B.code \parallel label(B.true) \parallel S_1.code \parallel \text{gen}(\text{goto } begin)$
$S \rightarrow S_1 S_2$	$S_1.next = newlabel(); \quad S_2.next = S.next$ $S.code = S_1.code \parallel label(S_1.next) \parallel S_2.code$

Table 8: SDD for flow-of-control statements (Dragon Book Fig. 6.36). $S.next$, $B.true$, and $B.false$ are inherited.

7.3 SDD for boolean expressions (jumping code)

How to remember the `||` and `&&` rules

- $B_1 \parallel B_2$: if B_1 is true we are done (go to $B.true$); if false, *test* B_2 , so $B_1.false$ is a new label placed in front of $B_2.code$.
- $B_1 \&\& B_2$: if B_1 is false we are done (go to $B.false$); if true, *test* B_2 , so $B_1.true$ is a new label in front of $B_2.code$.
- $!B_1$: simply *swap* the true and false labels.

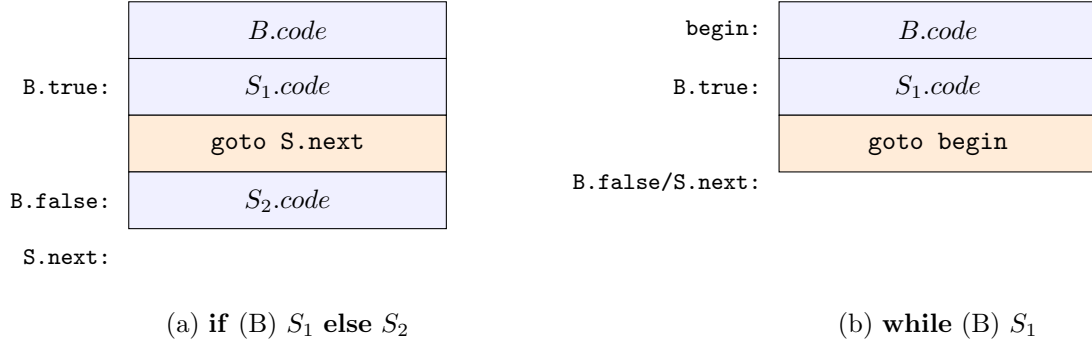


Figure 9: Layout of the generated code. *B.code* contains jumps to *B.true* or *B.false*; it “falls through” to neither, so the labels can be placed anywhere.

Production	Semantic rules
$B \rightarrow B_1 \parallel B_2$	$B_1.true = B.true; B_1.false = newlabel(); B_2.true = B.true; B_2.false = B.false$ $B.code = B_1.code \parallel \text{label}(B_1.false) \parallel B_2.code$
$B \rightarrow B_1 \&\& B_2$	$B_1.true = newlabel(); B_1.false = B.false; B_2.true = B.true; B_2.false = B.false$ $B.code = B_1.code \parallel \text{label}(B_1.true) \parallel B_2.code$
$B \rightarrow !B_1$	$B_1.true = B.false; B_1.false = B.true; B.code = B_1.code$
$B \rightarrow E_1 \text{ rel } E_2$	$B.code = E_1.code \parallel E_2.code \parallel \text{gen}(\text{if } E_1.addr \text{ rel.op } E_2.addr \text{ goto } B.true) \parallel \text{gen}(\text{goto } B.false)$
$B \rightarrow \text{true}$	$B.code = \text{gen}(\text{goto } B.true)$
$B \rightarrow \text{false}$	$B.code = \text{gen}(\text{goto } B.false)$

Table 9: SDD to generate jumping code for boolean expressions (Dragon Book Fig. 6.37).

Worked example: `if (x < 100 || x > 200 && x != y) x = 0;`

Step 1. $P \rightarrow S$: $S.next = L_1$.

Step 2. $S \rightarrow \text{if}(B)S_1$: $B.true = L_2$, $B.false = L_1$, $S_1.next = L_1$.

Step 3. $B = B_1 || B_2$, where $B_1 \equiv x < 100$ and $B_2 \equiv (x > 200 \ \&\& \ x \neq y)$:

- B_1 : $true = L_2$, $false = L_3$ (new).
- B_2 : $true = L_2$, $false = L_1$.

Step 4. $B_2 = B_3 \ \&\& \ B_4$ with $B_3 \equiv x > 200$, $B_4 \equiv x \neq y$:

- B_3 : $true = L_4$ (new), $false = L_1$.
- B_4 : $true = L_2$, $false = L_1$.

Step 5. Assemble: $B_1.code || L_3$: $B_2.code$, with $B_2.code = B_3.code || L_4$: $B_4.code$, followed by L_2 : $S_1.code$ and finally L_1 :

```

        if x < 100 goto L2
        goto L3
L3:     if x > 200 goto L4
        goto L1
L4:     if x != y goto L2
        goto L1
L2:     x = 0
L1:
```

Observe the redundant jumps (`goto L3` immediately followed by `L3:`) – a later optimisation phase or the *backpatching* method with fall-through can remove them.

Worked example: `while (a < b) { if (c < d) x = y + z; }`

```

L1:     if a < b goto L2           // begin / B.code
        goto L0                  // B.false = S.next
L2:     if c < d goto L3           // inner if: B.true = L3
        goto L1                  // inner S.next = begin = L1
L3:     t1 = y + z
        x = t1
        goto L1                  // goto begin
L0:                                     // S.next
```

7.4 Avoiding redundant gotos (fall-through)

Instead of

```

        if x > 200 goto L4
        goto L1
L4:     ...
```

we can generate the shorter form `ifFalse x > 200 goto L1` followed by the code at `L4`: if the true label is the next instruction (denote it by the special label *fall*), only one jump is needed. The rule for $B \rightarrow E_1 \text{ rel } E_2$ becomes: let $test = E_1.addr \text{ rel.op } E_2.addr$:

Condition	Generated jump code
$B.true \neq fall$ and $B.false \neq fall$	<code>if test goto B.true goto B.false</code>
$B.true \neq fall$ and $B.false = fall$	<code>if test goto B.true</code>
$B.true = fall$ and $B.false \neq fall$	<code>ifFalse test goto B.false</code>
both = <i>fall</i>	(no instruction)

8 Backpatching

8.1 The problem and the idea

In the one-pass, label-based scheme above we needed *inherited* labels ($B.true$, $S.next$) because a jump may be generated *before* its target is known. Inherited attributes make one-pass **bottom-up** translation awkward. **Backpatching** avoids this:

- When a jump is generated, its *target is left blank*.
- Each such incomplete jump is put on a **list** of jumps that will eventually go to the same target.
- When the target address becomes known, the instructions on the list are *patched* (filled in).

Thus we only need *synthesized* attributes (lists of instruction indices), and the whole translation can be done in a single bottom-up pass.

Backpatching primitives

- $makelist(i)$: create a new list containing only i , an index into the instruction array; return a pointer to it.
- $merge(p_1, p_2)$: concatenate the lists pointed to by p_1 and p_2 ; return the merged list.
- $backpatch(p, i)$: insert i as the target label for each instruction on the list p .
- $nextinstr$: global variable containing the index of the next instruction to be generated. $emit(\dots)$ appends an instruction and increments it.

8.2 Boolean expressions

Synthesized attributes: $B.truelist$ (instructions that jump when B is true) and $B.falselist$ (when false). A **marker nonterminal** $M \rightarrow \varepsilon$ records the index of the next instruction ($M.instr$).

Production	Semantic action
$B \rightarrow B_1 \parallel M B_2$	$\{ backpatch(B_1.falselist, M.instr);$ $B.truelist = merge(B_1.truelist, B_2.truelist); B.falselist = B_2.falselist; \}$
$B \rightarrow B_1 \&\& M B_2$	$\{ backpatch(B_1.truelist, M.instr);$ $B.truelist = B_2.truelist; B.falselist =$ $merge(B_1.falselist, B_2.falselist); \}$
$B \rightarrow ! B_1$	$\{ B.truelist = B_1.falselist; B.falselist = B_1.truelist; \}$
$B \rightarrow (B_1)$	$\{ B.truelist = B_1.truelist; B.falselist = B_1.falselist; \}$
$B \rightarrow E_1 \text{ rel } E_2$	$\{ B.truelist = makelist(nextinstr); B.falselist = makelist(nextinstr + 1);$ $emit(\text{if } E_1.addr \text{ rel.op } E_2.addr \text{ goto } -); emit(\text{goto } -); \}$
$B \rightarrow \text{true}$	$\{ B.truelist = makelist(nextinstr); emit(\text{goto } -); \}$
$B \rightarrow \text{false}$	$\{ B.falselist = makelist(nextinstr); emit(\text{goto } -); \}$
$M \rightarrow \varepsilon$	$\{ M.instr = nextinstr; \}$

Table 10: Translation scheme for boolean expressions using backpatching (Dragon Book Fig. 6.43).

Worked example: $x < 100 \parallel x > 200 \&\& x \neq y$ (instructions start at index 100)

Parse structure: $B \Rightarrow B_1 \parallel M_1 B_2$, where $B_2 \Rightarrow B_3 \&\& M_2 B_4$.

1. $B_1 \equiv x < 100$: emit 100: `if x<100 goto _, 101: goto _`
 $B_1.truelist = \{100\}$, $B_1.falselist = \{101\}$.
2. $M_1.instr = 102$.
3. $B_3 \equiv x > 200$: emit 102: `if x>200 goto _, 103: goto _`
 $B_3.truelist = \{102\}$, $B_3.falselist = \{103\}$.
4. $M_2.instr = 104$.
5. $B_4 \equiv x \neq y$: emit 104: `if x!=y goto _, 105: goto _`
 $B_4.truelist = \{104\}$, $B_4.falselist = \{105\}$.
6. Reduce $B_2 \rightarrow B_3 \&\& M_2 B_4$: $backpatch(\{102\}, 104) \Rightarrow$ instruction 102 becomes `if x>200 goto 104`.
 $B_2.truelist = \{104\}$, $B_2.falselist = merge(\{103\}, \{105\}) = \{103, 105\}$.
7. Reduce $B \rightarrow B_1 \parallel M_1 B_2$: $backpatch(\{101\}, 102) \Rightarrow$ instruction 101 becomes `goto 102`.
 $B.truelist = merge(\{100\}, \{104\}) = \{100, 104\}$, $B.falselist = \{103, 105\}$.

Result (still incomplete – the lists will be patched by the enclosing statement):

```

100: if x < 100 goto _      <- on truelist
101: goto 102
102: if x > 200 goto 104
103: goto _                <- on falselist
104: if x != y goto _      <- on truelist
105: goto _                <- on falselist

```

8.3 Flow-of-control statements

Synthesized attribute $S.nextlist$: list of jumps to the instruction that follows S . We use markers $M \rightarrow \varepsilon$ (records *nextinstr*) and $N \rightarrow \varepsilon$ (emits an incomplete `goto` and returns it on a list).

Grammar:

$$\begin{aligned}
 S &\rightarrow \text{if } (B) S_1 \mid \text{if } (B) S_1 \text{ else } S_2 \mid \text{while } (B) S_1 \mid \{ L \} \mid A; \\
 L &\rightarrow L_1 S \mid S
 \end{aligned}$$

Why the markers?

M remembers a position: “this is where the code for S_1 (or for the test) begins”. N generates the jump over the else-part and returns it for later patching. In the **while** rule the loop-back jump uses M_1 , which was recorded before $B.code$ was generated.

Worked example: `while (a < b) { if (c < d) x = y + z; }`

Assume numbering starts at 100; $M_1.instr = 100$.

1. $B \equiv a < b$: 100: `if a<b goto _, 101: goto _`; $B.truelist = \{100\}$, $B.falselist = \{101\}$.
2. $M_2.instr = 102$.
3. Inner $B' \equiv c < d$: 102: `if c<d goto _, 103: goto _`; $B'.truelist = \{102\}$, $B'.falselist = \{103\}$.
4. $M.instr = 104$; assignment $x = y + z$: 104: `t1 = y + z`, 105: `x = t1`;
 $S_1.nextlist = \text{null}$.
5. Inner **if**: $backpatch(\{102\}, 104)$; $S.nextlist = merge(\{103\}, \text{null}) = \{103\}$.

Production	Semantic action
$S \rightarrow \text{if } (B) \ M \ S_1$	$\{ \text{backpatch}(B.\text{truelist}, M.\text{instr}); \quad S.\text{nextlist} = \text{merge}(B.\text{falselist}, S_1.\text{nextlist}); \}$
$S \rightarrow \text{if } (B) \ M_1 \ S_1 \ N \ \text{else } M_2 \ S_2$	$\{ \text{backpatch}(B.\text{truelist}, M_1.\text{instr}); \text{backpatch}(B.\text{falselist}, M_2.\text{instr});$ $\quad \text{temp} = \text{merge}(S_1.\text{nextlist}, N.\text{nextlist}); \quad S.\text{nextlist} = \text{merge}(\text{temp}, S_2.\text{nextlist}); \}$
$S \rightarrow \text{while } M_1 (B) \ M_2 \ S_1$	$\{ \text{backpatch}(S_1.\text{nextlist}, M_1.\text{instr}); \text{backpatch}(B.\text{truelist}, M_2.\text{instr});$ $\quad S.\text{nextlist} = B.\text{falselist}; \text{emit}(\text{goto } M_1.\text{instr}); \}$
$S \rightarrow \{ L \}$	$\{ S.\text{nextlist} = L.\text{nextlist}; \}$
$S \rightarrow A;$	$\{ S.\text{nextlist} = \text{null}; \}$
$L \rightarrow L_1 \ M \ S$	$\{ \text{backpatch}(L_1.\text{nextlist}, M.\text{instr}); L.\text{nextlist} = S.\text{nextlist}; \}$
$L \rightarrow S$	$\{ L.\text{nextlist} = S.\text{nextlist}; \}$
$M \rightarrow \varepsilon$	$\{ M.\text{instr} = \text{nextinstr}; \}$
$N \rightarrow \varepsilon$	$\{ N.\text{nextlist} = \text{makelist}(\text{nextinstr}); \text{emit}(\text{goto } -); \}$

Table 11: Translation of statements with backpatching (Dragon Book Fig. 6.46).

6. **while:** $\text{backpatch}(\{103\}, 100)$ (i.e., 103: `goto 100`); $\text{backpatch}(\{100\}, 102)$ (i.e., 100: `if a<b goto 102`); $S.\text{nextlist} = B.\text{falselist} = \{101\}$; emit 106: `goto 100`.

```

100:  if a < b goto 102
101:  goto _          <- S.nextlist, patched by whoever follows
102:  if c < d goto 104
103:  goto 100
104:  t1 = y + z
105:  x = t1
106:  goto 100
107:  ...             <- (101 will be backpatched with 107)

```

Worked example: `if (a < b) x = 1; else x = 2;`

```

100:  if a < b goto 102    // B.truelist = {100}, patched to M1 = 102
101:  goto 104            // B.falselist = {101}, patched to M2 = 104
102:  x = 1
103:  goto _              // N: on S.nextlist
104:  x = 2
105:  ...                 // 103 is patched to 105 by the enclosing construct

```

8.4 Comparison: inherited labels vs. backpatching

	Label-based (Fig. 6.36/6.37)	SDD Backpatching (Fig. 6.43/6.46)
Attributes	Inherited labels (<i>true</i> , <i>false</i> , <i>next</i>)	Synthesized lists (<i>truelist</i> , <i>falselist</i> , <i>nextlist</i>)
Class of SDD	L-attributed	S-attributed (with markers <i>M</i> , <i>N</i>)
Best suited for	Top-down / recursive-descent (or AST traversal)	Bottom-up (LR) parsing, single pass
Code storage	Attribute <i>.code</i> , or incremental <i>gen</i>	Instruction array; patched later
Labels	Symbolic labels generated eagerly	Instruction indices known late

9 Switch Statements and Procedure Calls (Brief)

9.1 Switch statement

```
switch (E) {
    case V1: S1
    case V2: S2
    ...
    case V(n-1): S(n-1)
    default: Sn
}
```

Translation scheme: evaluate E into a temporary t , then a sequence of tests, followed by the code of each case.

```

    code to evaluate E into t
    goto test
L1:   code for S1
    goto next
L2:   code for S2
    goto next
...
Ln:   code for Sn
    goto next
test: if t = V1 goto L1
    if t = V2 goto L2
    ...
    if t = V(n-1) goto L(n-1)
    goto Ln
next:
```

Putting the tests *after* the case bodies lets the code generator recognise the test sequence and implement it with a **jump table** (for dense values) or binary search. The **case** instruction can also be used: `case t V1 L1 ... case t Vn Ln; case t t Ldefault.`

9.2 Procedure calls

For a call $id(E_1, E_2, \dots, E_n)$:

```

    code to evaluate E1 into t1
    code to evaluate E2 into t2
    ...
```

```

code to evaluate En into tn
param t1
param t2
...
param tn
call id, n          // or: y = call id, n    if a value is returned

```

Grammar and actions (the parameter addresses are collected in a queue, and then emitted as **param**):

$$\begin{aligned}
 S &\rightarrow \text{call id} (Elist) \quad \{ \text{for each } t \text{ in queue: } gen(\text{param } t); gen(\text{call id.addr}, size(queue)); \} \\
 Elist &\rightarrow Elist, E \quad \{ \text{append } E.addr \text{ to the end of } queue; \} \\
 Elist &\rightarrow E \quad \{ queue = [E.addr]; \}
 \end{aligned}$$

10 A Complete Worked Example

Translating a small program fragment to three-address code

Source fragment (all variables are `int`; `a` is `int a[10]`, so $w = 4$):

```

i = 0;
while (i < n && a[i] != key)
    i = i + 1;

```

Using the label-based SDD. $P \rightarrow S$ gives $S.next = L_0$. For the sequence $S_1 S_2$ (the assignment, then the loop) a label $L_1 = S_1.next$ is placed between them. For the loop, $begin = L_2$, $B.true = L_3$, $B.false = L_0$, and the body's $S_1.next = L_2$. For $B = B_1 \&\& B_2$ a new label $L_4 = B_1.true$ is created, with $B_1.false = L_0$, $B_2.true = L_3$, $B_2.false = L_0$.

```

        i = 0
L1:
L2:      if i < n goto L4      // B1 (begin of while)
        goto L0              // B1 false -> exit loop
L4:      t1 = i * 4           // B2: a[i] != key
        t2 = a[t1]
        if t2 != key goto L3
        goto L0              // B false -> exit loop
L3:      t3 = i + 1          // loop body
        i = t3
        goto L2              // back to begin
L0:

```

(L1 and L2 label the same instruction; a later pass can merge them.)

Using backpatching (instruction numbering from 100). Here $M_1.instr = 101$ (start of loop), the marker of `&&` gives 103, and $M_2.instr = 107$ (start of body).

```

100: i = 0
101: if i < n goto 103        // patched by the && marker
102: goto _                  // on B.falselist = S.nextlist
103: t1 = i * 4
104: t2 = a[t1]
105: if t2 != key goto 107    // truelist patched with M2 = 107
106: goto _                  // on B.falselist = S.nextlist
107: t3 = i + 1
108: i = t3
109: goto 101                // emitted by the while rule (M1)
110: ...                    // {102, 106} are backpatched with 110

```

Both of the incomplete jumps at 102 and 106 form $S.nextlist$ of the loop; whoever follows the

loop will patch them with the index of the next instruction, here 110.

11 Summary and Key Takeaways

- Intermediate code (syntax trees, DAGs, three-address code in quadruple/triple form) decouples front and back ends and enables machine-independent optimisation.
- Syntax alone does not give meaning. An **SDD** attaches *attributes* to grammar symbols and *semantic rules* to productions; **synthesized** attributes flow upward, **inherited** attributes flow downward or sideways. **S-attributed** $\subset$ **L-attributed**.
- An **SDT** embeds the actions in the grammar and fixes the order of evaluation. S-attributed SDDs $\rightarrow$ *postfix SDTs* (bottom-up). L-attributed SDDs can be implemented during top-down parsing, or, with marker nonterminals, during bottom-up parsing.
- Expressions: *E.addr* holds the result; *gen* emits TAC; temporaries are created by *newTemp*. Array references use offsets computed from the widths of sub-arrays.
- Boolean expressions are translated into **jumping code**, using the inherited labels *true/false*. Statements use *next*.
- **Backpatching** replaces inherited labels by synthesized lists (*makelist*, *merge*, *backpatch*), allowing one-pass bottom-up generation with markers *M* and *N*.

12 Practice Problems

- Q1.** Translate $x = a * (b + c) - d / (b + c)$ into (a) a syntax tree and DAG, (b) three-address code, (c) quadruples, triples, and indirect triples.
- Q2.** Using the SDD of Table 5, annotate the parse tree and generate TAC for $a = -(b + c) * (b + c)$.
- Q3.** For `int a[5][4]` (int width = 4), write the TAC for $x = a[i][j] + a[j][i]$. Give the width used at each dimension.
- Q4.** Generate jumping code (with inherited labels) for `if (a < b && !(c < d)) x = 1; else x = 2;`
- Q5.** Show, step by step, the *truelist/falselist* and the final code (with backpatching, starting at 100) for $a < b \parallel c < d \ \&\& \ e < f$.
- Q6.** Using backpatching, translate: `while (x < 10) { x = x + 1; if (x == 5) y = 0; }`
- Q7.** Classify as S-attributed, L-attributed, or neither: (a) $A \rightarrow BC$ with $C.i = B.s$ and $A.s = C.s$; (b) $A \rightarrow BC$ with $B.i = C.s$; (c) the SDD of Table 5.
- Q8.** Write the SDT for $E \rightarrow E_1 * E_2 \mid E_1 + E_2 \mid (E) \mid \text{id}$ that prints the *postfix* form, and show its output for $a + b * (c + d)$.

Answers to selected problems

Q1(b). The DAG shares $b + c$, so the TAC is

```

t1 = b + c
t2 = a * t1
t3 = d / t1
t4 = t2 - t3
x  = t4

```

Q3. Row width = $4 \times 4 = 16$; element width = 4.

```

t1 = i * 16
t2 = j * 4
t3 = t1 + t2
t4 = a[t3]
t5 = j * 16
t6 = i * 4
t7 = t5 + t6
t8 = a[t7]
t9 = t4 + t8
x  = t9

```

Q4. $B.true = L_2$, $B.false = L_3$, $S.next = L_1$; for B_1 & B_2 with $B_2 = !B_3$: $B_1.true = L_4$, $B_3.true = L_3$, $B_3.false = L_2$.

```

        if a < b goto L4
        goto L3
L4:      if c < d goto L3
        goto L2
L2:      x = 1
        goto L1
L3:      x = 2
L1:

```

Q7. (a) L-attributed ($C.i$ depends on the left sibling B) but not S-attributed. (b) Neither (inherited attribute of B depends on the right sibling C). (c) S-attributed (and hence L-attributed).

Q8. Postfix SDT: $E \rightarrow E_1 * E_2 \{print('*')\}$, $E \rightarrow E_1 + E_2 \{print('+')\}$, $E \rightarrow (E)$, $E \rightarrow id \{print(id.lexeme)\}$ (with precedence/associativity declared or via an unambiguous grammar). Output for $a + b * (c + d)$: $a \ b \ c \ d \ + \ * \ +$.

References

1. A. V. Aho, M. S. Lam, R. Sethi, J. D. Ullman, *Compilers: Principles, Techniques, and Tools*, 2nd ed., Pearson, 2007. – Chapter 5 (Syntax-Directed Translation) and Chapter 6 (Intermediate-Code Generation).